

7.4 达朗贝尔公式

定解问题

行波法

用行波法求解波动方程的基本思想：

先求出偏微分方程的通解，然后用定解条件确定特解。

评述：这一思想与常微分方程的解法是一样的。

关键步骤：通过变量变换，将波动方程化为便于积分的齐次二阶偏微分方程。

（一）波动方程的达朗贝尔公式

$$\left(\frac{\partial^2}{\partial t^2} - a^2 \frac{\partial^2}{\partial x^2}\right) u(x, t) = 0 \quad \longrightarrow \quad \frac{\partial^2}{\partial \xi \partial \eta} u(\xi, \eta) = 0$$

坐标变换

$$\left(\frac{\partial^2}{\partial t^2} - a^2 \frac{\partial^2}{\partial x^2}\right)u(x, t) = 0$$

$$a_{11} = 1, \quad a_{12} = 0, \quad a_{22} = -a^2 \quad a_{11} \left(\frac{dx}{dt}\right)^2 - 2a_{12} \left(\frac{dx}{dt}\right) + a_{22} = 0$$

$$\left(\frac{dx}{dt}\right)^2 - a^2 = 0$$

$$\begin{cases} \xi = x + at \\ \eta = x - at \end{cases} \longrightarrow \begin{cases} x = \frac{1}{2}(\xi + \eta) \\ t = \frac{1}{2a}(\xi - \eta) \end{cases} \quad u_{\xi\eta} = -\frac{1}{2A_{12}}[B_1 u_\xi + B_2 u_\eta + Cu + F]$$

$$\frac{\partial}{\partial \xi} = \frac{\partial t}{\partial \xi} \frac{\partial}{\partial t} + \frac{\partial x}{\partial \xi} \frac{\partial}{\partial x} = \frac{1}{2a} \frac{\partial}{\partial t} + \frac{1}{2} \frac{\partial}{\partial x} = \frac{1}{2a} \left(\frac{\partial}{\partial t} + a \frac{\partial}{\partial x} \right)$$

$$\frac{\partial}{\partial \eta} = \frac{\partial t}{\partial \eta} \frac{\partial}{\partial t} + \frac{\partial x}{\partial \eta} \frac{\partial}{\partial x} = -\frac{1}{2a} \frac{\partial}{\partial t} + \frac{1}{2} \frac{\partial}{\partial x} = -\frac{1}{2a} \left(\frac{\partial}{\partial t} - a \frac{\partial}{\partial x} \right)$$

$$\longrightarrow \boxed{-4a^2 \frac{\partial^2}{\partial \xi \partial \eta} u(\xi, \eta) = \left(\frac{\partial}{\partial t} + a \frac{\partial}{\partial x} \right) \left(\frac{\partial}{\partial t} - a \frac{\partial}{\partial x} \right) u(x, t) = 0} \quad 76$$

坐标变换 将 $\frac{\partial}{\partial x}$ 和 $\frac{\partial}{\partial t}$ 看作如同数一样的算子，可以进行加减乘除：

$$\left(\frac{\partial}{\partial t} + a \frac{\partial}{\partial x}\right) \left(\frac{\partial}{\partial t} - a \frac{\partial}{\partial x}\right) u(x, t) = 0$$

$$\begin{cases} \left(\frac{\partial}{\partial t} + a \frac{\partial}{\partial x}\right) u(x, t) = 0 & \frac{\partial u(x, t)}{\partial t} + a \frac{\partial u(x, t)}{\partial x} = 0 & \frac{1}{\partial t} + a \frac{1}{\partial x} = 0 \\ \left(\frac{\partial}{\partial t} - a \frac{\partial}{\partial x}\right) u(x, t) = 0 & \frac{\partial u(x, t)}{\partial t} - a \frac{\partial u(x, t)}{\partial x} = 0 & \frac{1}{\partial t} - a \frac{1}{\partial x} = 0 \end{cases}$$

$$\begin{cases} \xi = x + at \\ \eta = x - at \end{cases} \quad \longrightarrow \quad \begin{cases} x = \frac{1}{2}(\xi + \eta) \\ t = \frac{1}{2a}(\xi - \eta) \end{cases}$$

$$\frac{\partial}{\partial \xi} = \frac{\partial t}{\partial \xi} \frac{\partial}{\partial t} + \frac{\partial x}{\partial \xi} \frac{\partial}{\partial x} = \frac{1}{2a} \frac{\partial}{\partial t} + \frac{1}{2} \frac{\partial}{\partial x} = \frac{1}{2a} \left(\frac{\partial}{\partial t} + a \frac{\partial}{\partial x} \right)$$

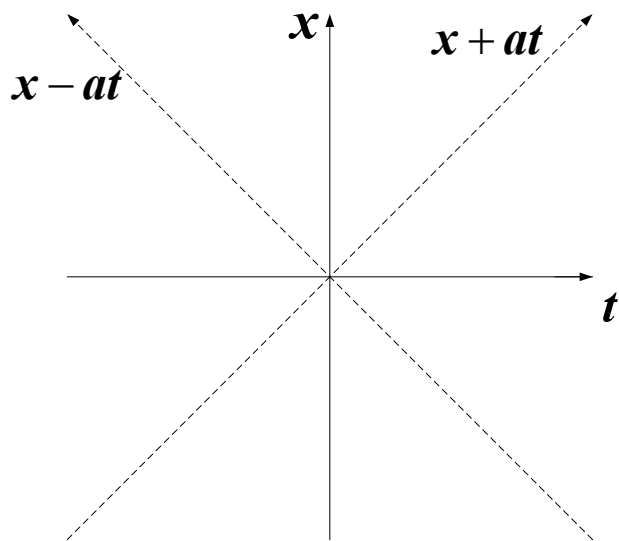
$$\frac{\partial}{\partial \eta} = \frac{\partial t}{\partial \eta} \frac{\partial}{\partial t} + \frac{\partial x}{\partial \eta} \frac{\partial}{\partial x} = -\frac{1}{2a} \frac{\partial}{\partial t} + \frac{1}{2} \frac{\partial}{\partial x} = -\frac{1}{2a} \left(\frac{\partial}{\partial t} - a \frac{\partial}{\partial x} \right)$$

$$\longrightarrow \boxed{-4a^2 \frac{\partial^2}{\partial \xi \partial \eta} u(\xi, \eta) = \left(\frac{\partial}{\partial t} + a \frac{\partial}{\partial x}\right) \left(\frac{\partial}{\partial t} - a \frac{\partial}{\partial x}\right) u(x, t) = 0} \quad 77$$

坐标变换

将 $\frac{\partial}{\partial x}$ 和 $\frac{\partial}{\partial t}$ 看作如同数一样的算子，可以进行加减乘除：

$$\left(\frac{\partial}{\partial t} + a \frac{\partial}{\partial x}\right) \left(\frac{\partial}{\partial t} - a \frac{\partial}{\partial x}\right) u(x, t) = 0$$



当 $a=1$ ，相当于沿 x 和 t 求导，变成沿对角线求导。当 a 不为1，则求导的线进行相应的角度变化。

变换： $x = \frac{1}{2}(\xi + \eta)$ 和 $t = \frac{1}{2a}(\xi - \eta)$
显然， $\xi = x + at$ $\eta = x - at$

$$\frac{\partial}{\partial \xi} = \frac{\partial t}{\partial \xi} \frac{\partial}{\partial t} + \frac{\partial x}{\partial \xi} \frac{\partial}{\partial x} = \frac{1}{2a} \frac{\partial}{\partial t} + \frac{1}{2} \frac{\partial}{\partial x} = \frac{1}{2a} \left(\frac{\partial}{\partial t} + a \frac{\partial}{\partial x} \right)$$

$$\frac{\partial}{\partial \eta} = \frac{\partial t}{\partial \eta} \frac{\partial}{\partial t} + \frac{\partial x}{\partial \eta} \frac{\partial}{\partial x} = -\frac{1}{2a} \frac{\partial}{\partial t} + \frac{1}{2} \frac{\partial}{\partial x} = -\frac{1}{2a} \left(\frac{\partial}{\partial t} - a \frac{\partial}{\partial x} \right)$$



$$-4a^2 \frac{\partial^2}{\partial \xi \partial \eta} u(\xi, \eta) = 0$$

$$\left(\frac{\partial^2}{\partial t^2} - a^2 \frac{\partial^2}{\partial x^2}\right)u(x, t) = 0 \quad \longrightarrow \quad \frac{\partial^2}{\partial \xi \partial \eta} u(\xi, \eta) = 0$$

(1) 通解 对 η 积分: $\frac{\partial}{\partial \xi} u(x, t) = f(\xi)$ 积分常数依赖于 ξ

再对 ξ 积分: $u = \int f(\xi) d\xi + f_2(\eta) = \underbrace{f_1(\xi)} + \underbrace{f_2(\eta)}$

$$u(x, t) = f_1(x + at) + f_2(x - at)$$

以 f_2 为例讨论其意义

作坐标变换: $\begin{cases} X = x - at \\ T = t \end{cases} \quad f_2(x - at) = f_2(X)$

新坐标的时间与旧坐标同, 新坐标的原点 $X=0$ 在旧坐标中以速度 a 运动; 函数的图像在动坐标系中保持不变。

$f_2(x-at)$ 是以速度 a 沿 x 轴正方向运动的行波,
 $f_1(x+at)$ 是以速度 a 沿 x 轴反方向运动的行波。

(2) 达朗贝尔公式

$$u(x, t) = f_1(x + at) + f_2(x - at)$$

确定待定函数的形式

无限长，即无边界条件

设初始条件为 $u|_{t=0} = \varphi(x)$ 和 $u_t|_{t=0} = \psi(x)$ $(-\infty < x < \infty)$

$$\begin{cases} f_1(x) + f_2(x) = \varphi(x) \\ af_1'(x) - af_2'(x) = \psi(x) \end{cases} \Rightarrow f_1(x) - f_2(x) = \frac{1}{a} \int_{x_0}^x \psi(\xi) d\xi + f_1(x_0) - f_2(x_0)$$

$$f_1(x) = \frac{1}{2} \varphi(x) + \frac{1}{2a} \int_{x_0}^x \psi(\xi) d\xi + \frac{1}{2} [f_1(x_0) - f_2(x_0)]$$

$$\left. \begin{aligned} f_1(x + at) &= \frac{1}{2} \varphi(x + at) + \frac{1}{2a} \int_{x_0}^{x+at} \psi(\xi) d\xi + \frac{1}{2} [f_1(x_0) - f_2(x_0)] \\ f_2(x - at) &= \frac{1}{2} \varphi(x - at) - \frac{1}{2a} \int_{x_0}^{x-at} \psi(\xi) d\xi - \frac{1}{2} [f_1(x_0) - f_2(x_0)] \end{aligned} \right\} \rightarrow u = f_1(x + at) + f_2(x - at)$$

$$u(x, t) = \frac{1}{2} [\varphi(x + at) + \varphi(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi$$

$$u(x, t) = \frac{1}{2}[\varphi(x + at) + \varphi(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi$$

从达朗贝尔公式可以看出，波动方程的解，是初始条件的演化。方程本身并不可能产生出超出初始条件的，额外的形式来。

而这种演化又受到边界条件的限制。

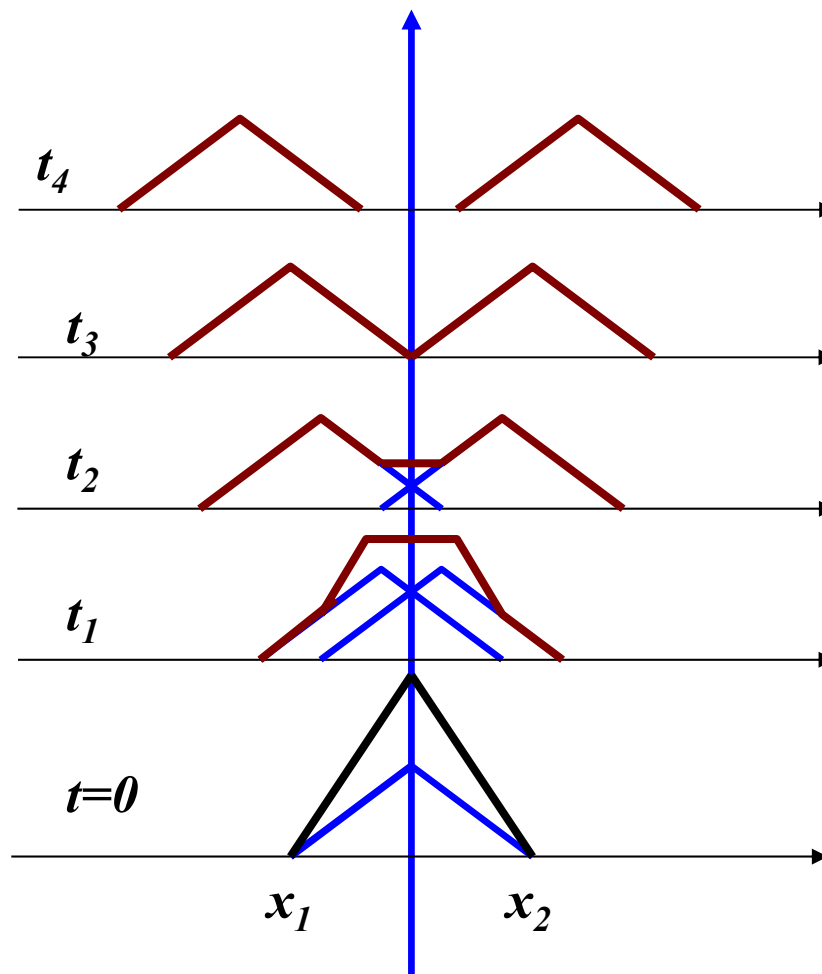
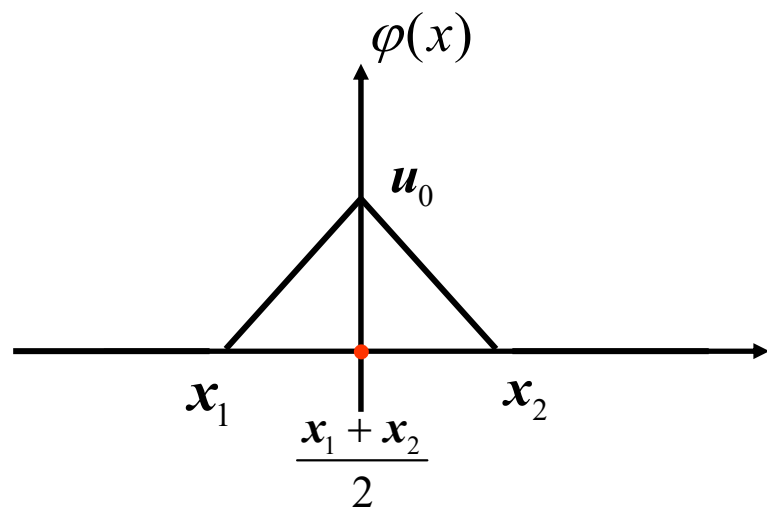
这就说明了初始条件和边界条件在确定波动方程的解时的重要性。

达朗贝尔解表示沿 x 轴正、反向传播的两列波的叠加

设初速度为零

$$u_t|_{t=0} = \psi(x) = 0$$

$$u|_{t=0} = \varphi(x) = \begin{cases} 2u_0 \frac{x-x_1}{x_2-x_1}, & x_1 \leq x \leq \frac{x_1+x_2}{2} \\ 2u_0 \frac{x_2-x}{x_2-x_1}, & \frac{x_1+x_2}{2} \leq x \leq x_2 \\ 0 & x < x_1, \quad x > x_2 \end{cases}$$



由达朗贝尔公式

$$u(x,t) = \frac{1}{2} [\varphi(x+at) + \varphi(x-at)]$$

设初位移为零

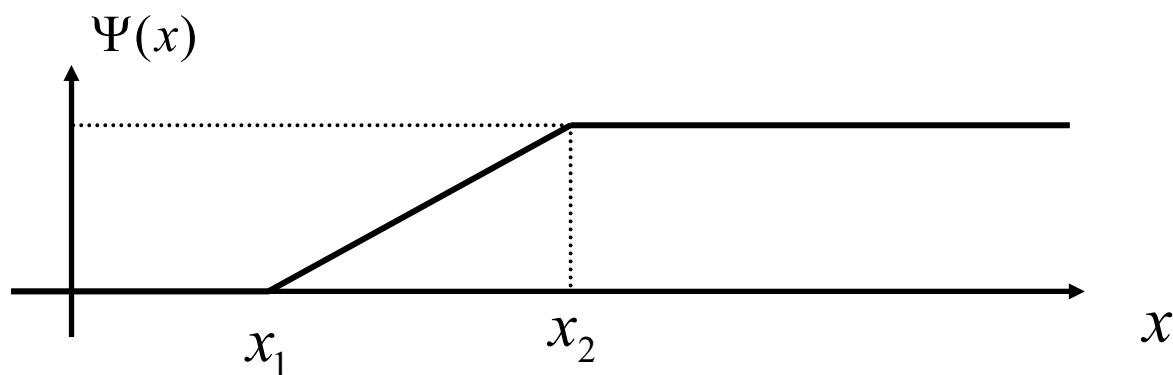
假使初始速度在区间 $[x_1, x_2]$ 上是常数 ψ_0

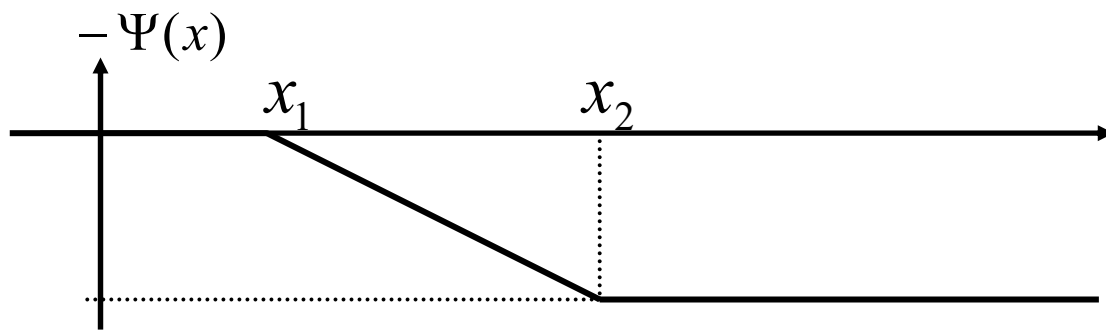
$$u|_{t=0} = \varphi(x) = 0$$

$$u_t|_{t=0} = \psi(x) = \begin{cases} \psi_0 & x \in (x_1, x_2) \\ 0 & x \notin (x_1, x_2) \end{cases}$$

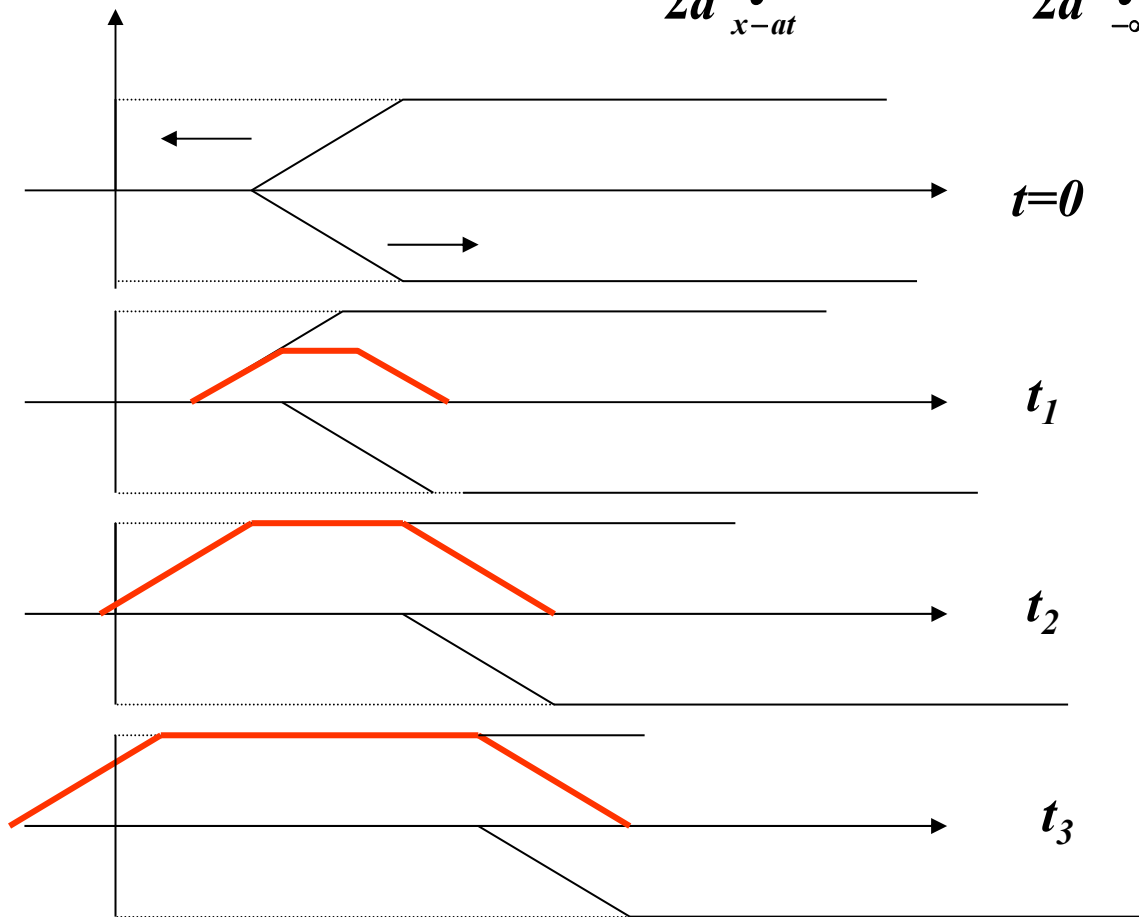
解:
$$u(x, t) = \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi = \frac{1}{2a} \int_{-\infty}^{x+at} \psi(\xi) d\xi - \frac{1}{2a} \int_{-\infty}^{x-at} \psi(\xi) d\xi \quad \text{其中}$$

$$\Psi(x) = \frac{1}{2a} \int_{-\infty}^x \psi(\xi) d\xi$$





$$u(x, t) = \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi = \frac{1}{2a} \int_{-\infty}^{x+at} \psi(\xi) d\xi - \frac{1}{2a} \int_{-\infty}^{x-at} \psi(\xi) d\xi$$



(二) 端点的反射

半无限长弦的自由振动

$$\left\{ \begin{array}{l} (\frac{\partial^2}{\partial t^2} - a^2 \frac{\partial^2}{\partial x^2})u(x, t) = 0 \quad (0 < x < \infty) \quad \text{一个端点固定} \\ \text{设初始条件为 } u|_{t=0} = \varphi(x) \text{ 和 } u_t|_{t=0} = \psi(x) \quad (0 < x < \infty) \\ \text{第一类边界条件 } u|_{x=0} = 0 \end{array} \right.$$

达朗贝尔公式是无限长弦的公式。由于自变量限制为 $x \geq 0$

$$u(x, t) = \frac{1}{2} [\varphi(x + at) + \varphi(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi$$

$t > x/a$ 时, 上式后两项无意义, 必须将 $u(x, t)$ 延拓到这个范围

$u(0, t) = 0$ 做奇延拓: $\varphi(x) \rightarrow \Phi(x)$ $\psi(x) \rightarrow \Psi(x)$

$$\Phi(x) = \begin{cases} \varphi(x) & (x \geq 0) \\ -\varphi(-x) & (x < 0) \end{cases} \quad \Psi(x) = \begin{cases} \psi(x) & (x \geq 0) \\ -\psi(-x) & (x < 0) \end{cases} \quad 85$$

$$u(x, t) = \frac{1}{2}[\Phi(x + at) + \Phi(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} \Psi(\xi) d\xi$$

$$u = \frac{1}{2}[\varphi(x + at) + \varphi(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi \quad (t \leq \frac{x}{a})$$

$$x \leq at \text{ 即 } t \geq \frac{x}{a}$$

$$u = \frac{1}{2}[\Phi(x + at) + \Phi(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} \Psi(\xi) d\xi$$

$$= \frac{1}{2}[\varphi(x + at) - \varphi(at - x)] + \frac{1}{2a} \int_0^{x+at} \psi(\xi) d\xi$$

$$+ \frac{1}{2a} \int_{x-at}^0 [-\psi(-\xi)] d\xi$$

$$\frac{1}{2}[\varphi(x+at)-\varphi(at-x)]+\frac{1}{2a}\int_0^{x+at}\psi(\xi)d\xi$$

$$+\frac{1}{2a}\int_{x-at}^0[-\psi(-\xi)]d\xi$$

$$\frac{1}{2a}\int_{x-at}^0[-\psi(-\xi)]d\xi=\frac{1}{2a}\int_{at-x}^0\psi(\xi)d\xi=-\frac{1}{2a}\int_0^{at-x}\psi(\xi)d\xi$$

$$u=\frac{1}{2}[\varphi(x+at)-\varphi(at-x)]+\frac{1}{2a}\int_0^{x+at}\psi(\xi)d\xi$$

$$-\frac{1}{2a}\int_0^{at-x}\psi(\xi)d\xi$$

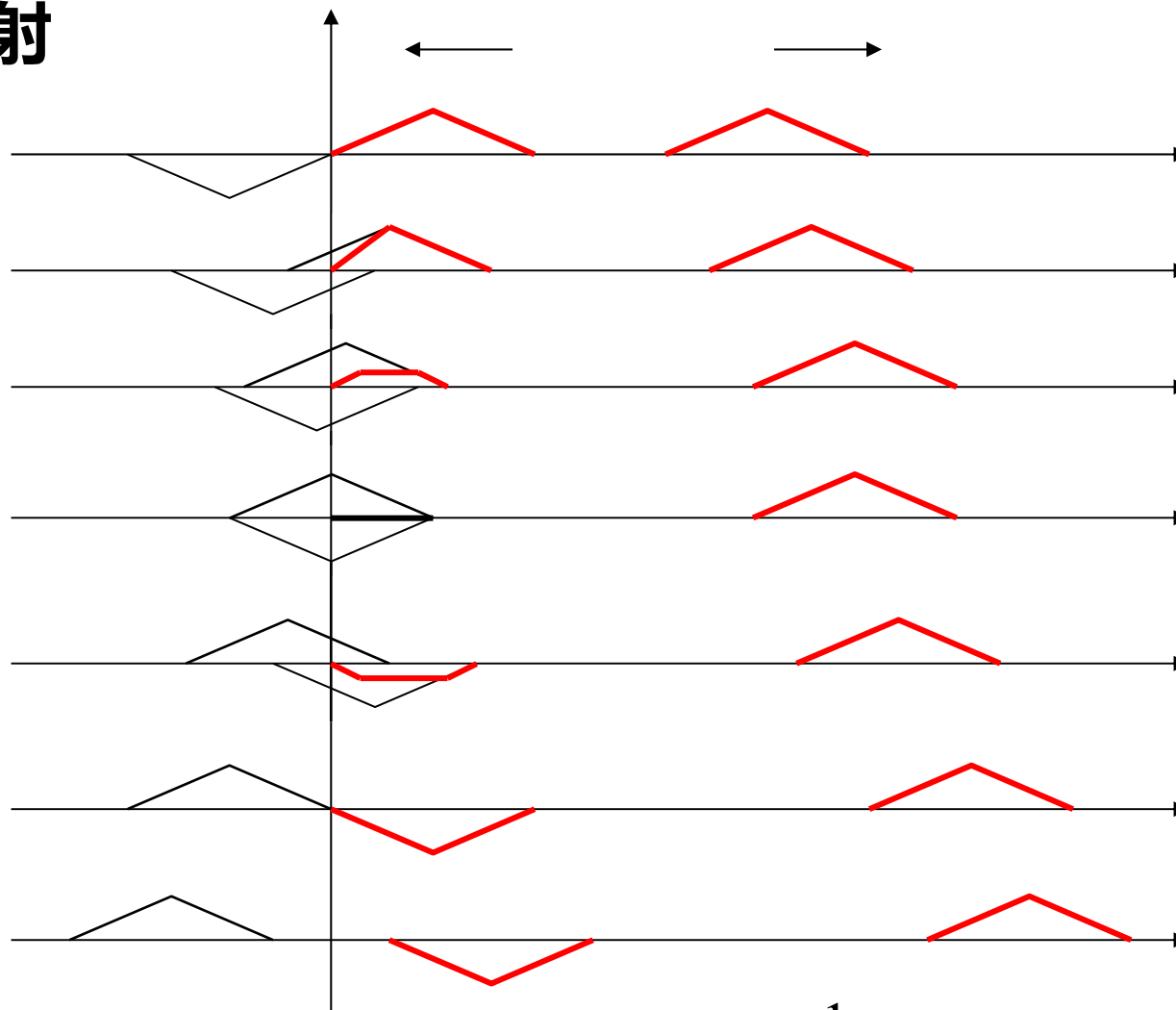
$$u = \frac{1}{2}[\varphi(x+at) - \varphi(at-x)] + \frac{1}{2a} \int_0^{x+at} \psi(\xi) d\xi \\ - \frac{1}{2a} \int_0^{at-x} \psi(\xi) d\xi$$

$$= \frac{1}{2}[\varphi(x+at) - \varphi(at-x)] + \frac{1}{2a} \int_{at-x}^{x+at} \psi(\xi) d\xi$$

$$(t \geq \frac{x}{a})$$

$$u = \frac{1}{2}[\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi \quad (t \leq \frac{x}{a})$$

开始反射



半波损失

只有初始位移，没有初始速度

$$u = \frac{1}{2}[\varphi(x+at) + \varphi(x-at)] \quad (t \leq \frac{x}{a})$$

$$u = \frac{1}{2}[\varphi(x+at) - \varphi(at-x)] \quad (t \geq \frac{x}{a})$$

$$\left\{ \begin{array}{l} \text{一个端点自由} \quad \left(\frac{\partial^2}{\partial t^2} - a^2 \frac{\partial^2}{\partial x^2} \right) u(x, t) = 0 \quad (0 < x < \infty) \\ \text{设初始条件为} \quad u|_{t=0} = \varphi(x) \quad \text{和} \quad u_t|_{t=0} = \psi(x) \\ \text{第二类边界条件} \quad u_x|_{x=0} = 0 \end{array} \right.$$

做偶延拓

$$\Phi(x) = \begin{cases} \varphi(x) & (x \geq 0) \\ \varphi(-x) & (x < 0) \end{cases} \quad \Psi(x) = \begin{cases} \psi(x) & (x \geq 0) \\ \psi(-x) & (x < 0) \end{cases}$$

$$u(x, t) = \frac{1}{2} [\Phi(x + at) + \Phi(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} \Psi(\xi) d\xi$$

$$u = \frac{1}{2}[\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi \quad (t \leq \frac{x}{a})$$

$$x \leq at \text{ 即 } t \geq \frac{x}{a}$$

$$\begin{aligned} u &= \frac{1}{2}[\Phi(x+at) + \Phi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \Psi(\xi) d\xi \\ &= \frac{1}{2}[\varphi(x+at) + \varphi(at-x)] + \frac{1}{2a} \int_0^{x+at} \psi(\xi) d\xi \\ &\quad + \frac{1}{2a} \int_{x-at}^0 \psi(-\xi) d\xi \end{aligned}$$

$$\frac{1}{2a} \int_{x-at}^0 \psi(-\xi) d\xi = -\frac{1}{2a} \int_{at-x}^0 \psi(\xi) d\xi = \frac{1}{2a} \int_0^{at-x} \psi(\xi) d\xi$$

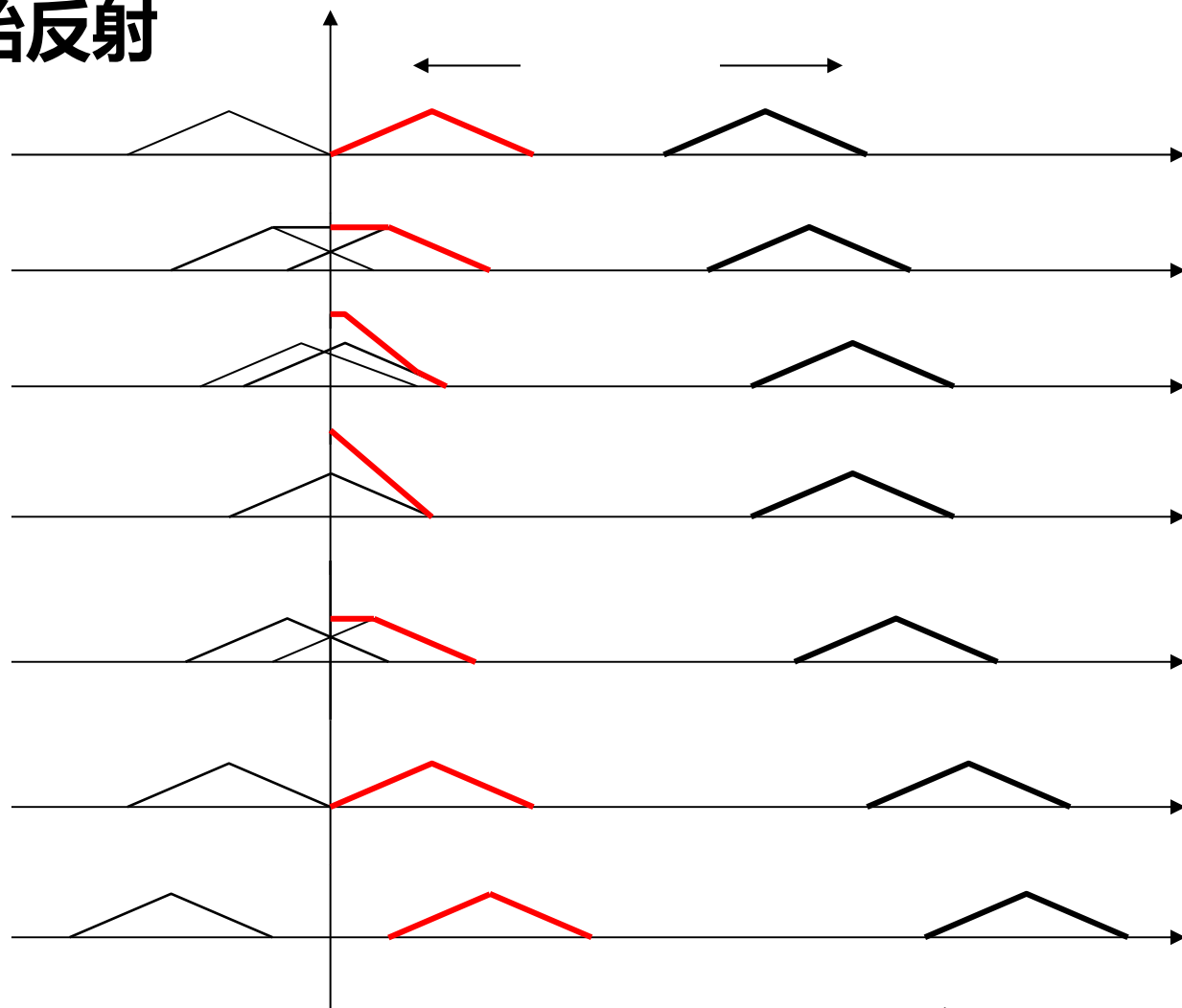
$$\frac{1}{2a} \int_{x-at}^0 \psi(-\xi) d\xi = \frac{1}{2a} \int_0^{at-x} \psi(\xi) d\xi$$

$$u = \frac{1}{2} [\varphi(x+at) + \varphi(at-x)] + \frac{1}{2a} \int_0^{x+at} \psi(\xi) d\xi$$

$$+ \frac{1}{2a} \int_0^{at-x} \psi(\xi) d\xi \quad \left(t \geq \frac{x}{a}\right)$$

$$u = \frac{1}{2} [\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi \quad \left(t \leq \frac{x}{a}\right)$$

开始反射



无半波损失

只有初始位移，没有初始速度

$$u = \frac{1}{2} [\varphi(x+at) + \varphi(x-at)] \quad (t \leq \frac{x}{a})$$

$$u = \frac{1}{2} [\varphi(x+at) + \varphi(at-x)] \quad (t \geq \frac{x}{a})$$

(三) 定解问题的适定性

定解问题的解:

() 存在性; () 唯一性; () 稳定性

满足上述条件的定解问题, 称为是**适定的**。

1、达朗贝尔解的存在性

$$u(x,t) = \frac{1}{2}[\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi$$

$\varphi(x)$ **有二阶导数**, $\psi(x)$ **有一阶导数**, 达朗贝尔解存在

2、达朗贝尔解的稳定性

考虑初始条件有两组，差别微小

$$u(x, t)|_{t=0} = \begin{cases} \varphi_1(x) \\ \varphi_2(x) \end{cases}$$

$$u_t(x, t)|_{t=0} = \begin{cases} \psi_1(x) \\ \psi_2(x) \end{cases}$$

$$|\varphi_1(x) - \varphi_2(x)| < \delta$$

$$|\psi_1(x) - \psi_2(x)| < \delta$$

$$u = \frac{1}{2}[\varphi(x + at) + \varphi(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi$$

$$u = \frac{1}{2}[\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi$$

$$u(x,t)|_{t=0} = \begin{cases} \varphi_1(x) \\ \varphi_2(x) \end{cases} \quad u_t(x,t)|_{t=0} = \begin{cases} \psi_1(x) \\ \psi_2(x) \end{cases}$$

$$\begin{aligned} |u_1 - u_2| &\leq \frac{1}{2} |\varphi_1(x+at) - \varphi_2(x+at)| \\ &\quad + \frac{1}{2} |\varphi_1(x-at) - \varphi_2(x-at)| \quad |\varphi_1(x) - \varphi_2(x)| < \delta \\ &\quad + \frac{1}{2a} \int_{x-at}^{x+at} |\psi_1(\xi) - \psi_2(\xi)| d\xi \quad |\psi_1(x) - \psi_2(x)| < \delta \\ &< \frac{1}{2} \delta + \frac{1}{2} \delta + \frac{1}{2a} \int_{x-at}^{x+at} \delta d\xi < \delta (1+t) \end{aligned}$$

达朗贝尔解的稳定

(四) 跃变点的反射

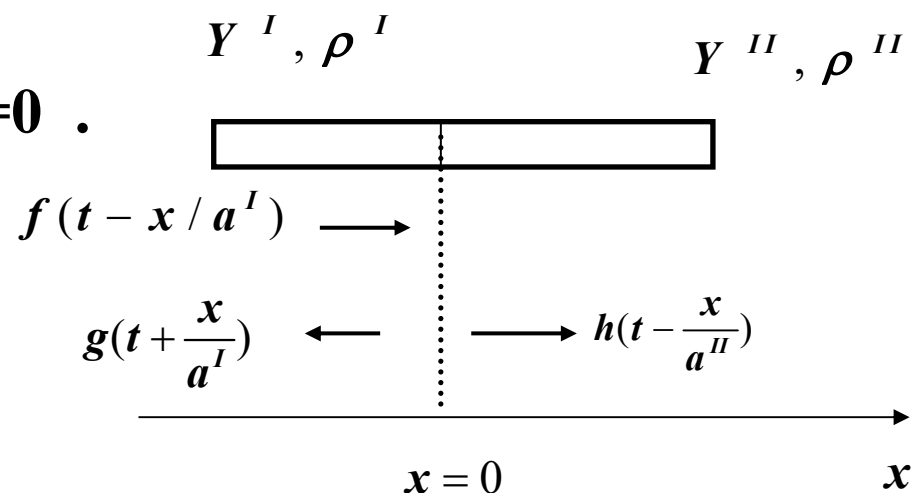
无限长杆, $x < 0$, $x > 0$ 两部分的杨氏模量和密度分别为 $Y^I, \rho^I, Y^{II}, \rho^{II}$. $x=0$ 是跃变点。

设有行波 $u(x, t) = f(t - x/a^I)$ 从区域 I 向 $x=0$ 点运动。到 $x=0$ 产生反射和透射。

取此波在 $t=0$ 时刻抵达 $x=0$.

$$\begin{cases} u_{tt}^I - (a^I)^2 u_{xx}^I = 0, & (x < 0) \\ u^I \Big|_{t \leq 0} = f\left(t - \frac{x}{a^I}\right) & (x < 0) \end{cases}$$

$$\begin{cases} u_{tt}^{II} - (a^{II})^2 u_{xx}^{II} = 0, & (x > 0) \\ u^{II} \Big|_{t \leq 0} = 0, u_t^{II} \Big|_{t \leq 0} = 0 & (x > 0) \end{cases}$$



$$a^I = \frac{Y^I}{\rho^I} \quad a^{II} = \frac{Y^{II}}{\rho^{II}}$$

衔接条件

$$\begin{cases} u^I \Big|_{x=0} = u^{II} \Big|_{x=0}, \\ Y^I u_x^I \Big|_{x=0} = Y^{II} u_x^{II} \Big|_{x=0}, \end{cases}$$

区域 I 中的行波: $u^I(x, t) = f(t - \frac{x}{a^I}) + g(t + \frac{x}{a^I}), \quad (x < 0, t > 0)$

$$t = 0 \quad g(\frac{x}{a^I}) = 0 \quad \Rightarrow \quad \boxed{g(\xi) = 0, g'(\xi) = 0 \quad (\xi < 0)}$$

区域 II 中, 只有透射波 $u^{II}(x, t) = h(t - \frac{x}{a^{II}}) \quad (x > 0, t > 0)$

$$t = 0 \quad h(-\frac{x}{a^{II}}) = 0, h'(-\frac{x}{a^{II}}) = 0 \quad \Rightarrow \quad \boxed{h(\eta) = 0, h'(\eta) = 0 \quad (\eta < 0)}$$

衔接条件
$$\begin{cases} f(t) + g(t) = h(t), \\ -\frac{1}{a^I} Y^I [f'(t) - g'(t)] = -\frac{1}{a^{II}} Y^{II} h'(t) \end{cases} \quad (t > 0)$$

$$\begin{cases} f(t) + g(t) = h(t), \\ a^{II} Y^I [f(t) - g(t)] = a^I Y^{II} h(t) \end{cases} \quad (t > 0)$$

$$\begin{cases} h(\xi) = \frac{2a^{II} Y^I}{a^{II} Y^I + a^I Y^{II}} f(\xi) \\ g(\eta) = \frac{a^{II} Y^I - a^I Y^{II}}{a^{II} Y^I + a^I Y^{II}} f(\eta) \end{cases} \quad (\xi > 0, \eta > 0)$$

$$h\left(t - \frac{x}{a''}\right) = \begin{cases} 0 & (t \leq \frac{x}{a''}) \\ \frac{2a''Y^I}{a''Y^I + a^IY^{II}} f\left(t - \frac{x}{a''}\right) & (t > \frac{x}{a''}) \end{cases}$$

$$g\left(t + \frac{x}{a^I}\right) = \begin{cases} 0 & (t \leq \frac{-x}{a^I}) \\ \frac{a''Y^I - a^IY^{II}}{a''Y^I + a^IY^{II}} f\left(t + \frac{x}{a^I}\right) & (t > \frac{-x}{a^I}) \end{cases}$$

$$\text{又} \quad \left| \frac{a''Y^I - a^IY^{II}}{a''Y^I + a^IY^{II}} \right|^2 + \left| \frac{2a''Y^I}{a''Y^I + a^IY^{II}} \right|^2 = 1$$

$$\left| \frac{a''Y^I - a^IY^{II}}{a''Y^I + a^IY^{II}} \right|^2$$

反射系数

$$\left| \frac{2a''Y^I}{a''Y^I + a^IY^{II}} \right|^2$$

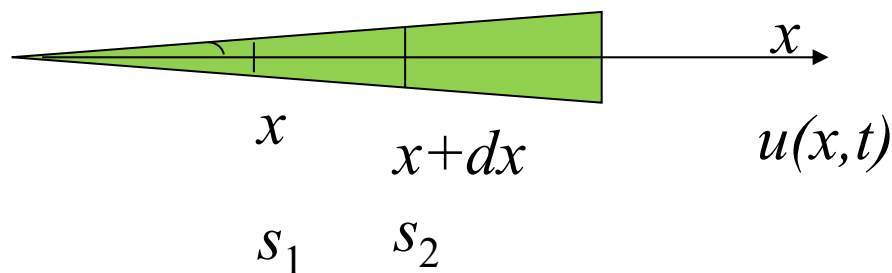
透射系数

习题 7.1.2

用匀质材料制做细圆锥杆，试推导它的纵振动方程。

解：如图选坐标系，选 dx 段为研究对象， dx 段两边受拉力分别为

$$YS_1 \frac{\partial u}{\partial x} \Big|_x \quad YS_2 \frac{\partial u}{\partial x} \Big|_{x+dx}$$



由牛顿第二定律： $Y(S_2 u_x \Big|_{x+dx} - S_1 u_x \Big|_x) = \rho S_1 dx u_{tt}$

$$Y \frac{S_2 u_x \Big|_{x+dx} - S_1 u_x \Big|_x}{dx} = \rho S_1 u_{tt}$$

$$u_{tt} - \frac{a^2}{x^2} \frac{\partial}{\partial x} (x^2 u_x) = 0 \quad a = \sqrt{\frac{Y}{\rho}}$$

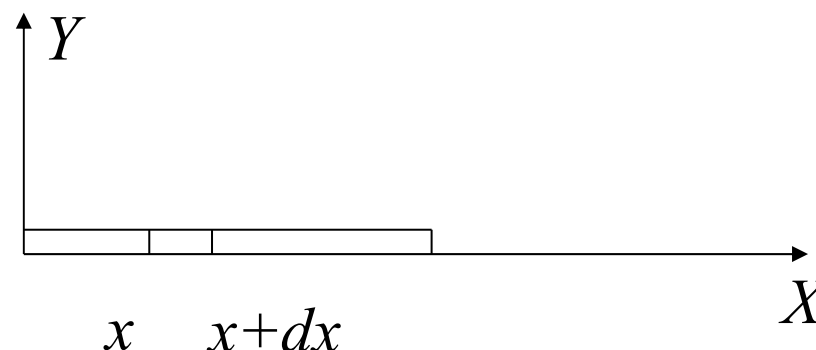
习题7.1.7: 长为 l 的柔软均质绳索, 一端固定在以匀速转动的竖直轴上, 由于惯性离心力的作用, 这弦的平衡位置应是水平线。试推导此绳相对于水平线的横振动方程。

解: 如图选坐标系, 由于惯性离心力的作用, 绳内各处受力不同, x 处的拉力为

$$T(x) = \int_x^l \omega^2 x \rho dx = \frac{1}{2} \rho \omega^2 (l^2 - x^2)$$

$$T_2 u_x|_{x+dx} - T_1 u_x|_x = \rho dx u_{tt}$$

$$\frac{\partial}{\partial x} T u_x = \rho u_{tt}$$

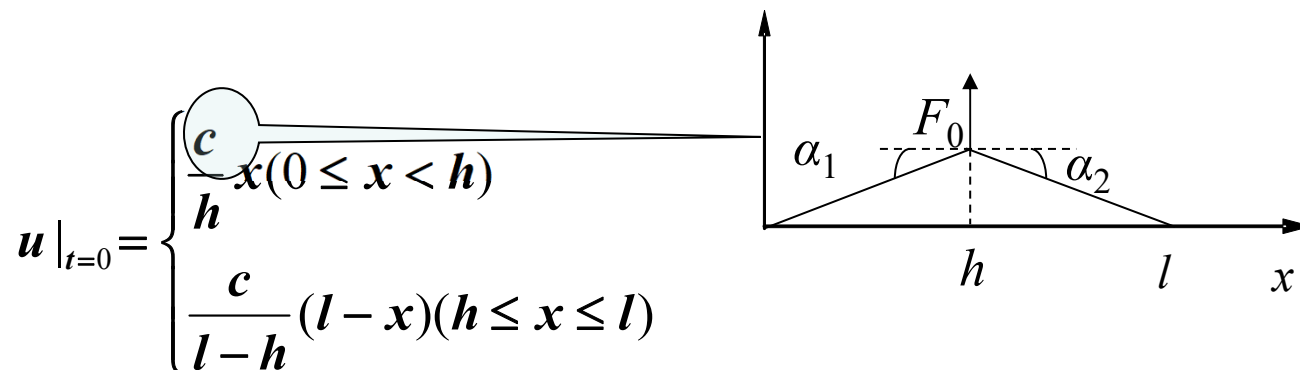


$$\frac{\partial}{\partial x} \left[\frac{1}{2} \rho \omega^2 (l^2 - x^2) u_x \right] = \rho u_{tt}$$

即
$$u_{tt} - \frac{1}{2} \omega^2 \frac{\partial}{\partial x} [(l^2 - x^2) u_x] = 0$$

习题7.2.1 如右图，在 h 处受到拉力 F_0 ，写出初始位移。

解



确定 c :

$$\sin a_1 = \tan a_1 = \frac{c}{h} \quad (1)$$

$$\sin a_2 = \tan a_2 = \frac{c}{l-h} \quad (2)$$

$$\cos a_1 = \cos a_2 \approx 1 \quad ds=dx$$

力平衡条件:

$$\begin{cases} F_0 - T_1 \sin a_1 - T_2 \sin a_2 = 0 & (3) \\ T_2 \cos a_2 - T_1 \cos a_1 = 0 & (4) \end{cases}$$

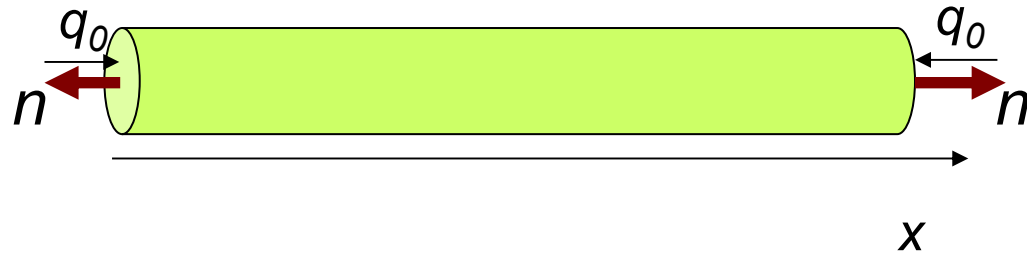
解出: $c = \frac{F_0 h(l-h)}{T_0 l}$

$$u|_{t=0} = \begin{cases} \frac{F_0(l-h)}{T_0 l} x & (0 < x < h) \\ \frac{F_0 h}{T_0 l} (l-x) & (h < x < l) \end{cases}$$

习题7.2.3 长为 l 的均匀杆，两端有恒定热流进入，其强度为 q_0 ，写出这个热传导问题的边界条件。

解：

在边界上有：



$$q_0 = -k \frac{\partial u}{\partial n}$$

$$x=l \text{ 处: } -k \frac{\partial u}{\partial n} \Big|_{x=l} = -k \frac{\partial u}{\partial x} \Big|_{x=l} = q_n = -q_0 \rightarrow \frac{\partial u}{\partial x} \Big|_{x=l} = \frac{q_0}{k}$$

$$x=0 \text{ 处: } -k \frac{\partial u}{\partial n} \Big|_{x=0} = -k \frac{\partial u}{\partial (x)} \Big|_{x=0} = q_n = q_0 \rightarrow \frac{\partial u}{\partial x} \Big|_{x=0} = -\frac{q_0}{k}$$

若端点是绝热的，则 $\frac{\partial u}{\partial x} \Big|_{x=l} = \frac{\partial u}{\partial x} \Big|_{x=0} = 0$